```
[['dbfs:/Volumes/ml_in_action/rag_chatbot/files/raw_documents/2303.10130.pdf',
  '7 Conclusion\n'
  'In conclusion, this study offers an examination of the potential impact of LLMs on various '
  'occupations and industries within the U.S. economy. By applying a new rubric for understanding '
  'LLM capabilities and their potential effects on jobs, we have observed that most occupations '
  'exhibit some degree of exposure to LLMs, with higher-wage occupations generally presenting more '
  'tasks with high exposure. Our analysis indicates that approximately 19% of jobs have at least '
  '50% of their tasks exposed to LLMs when considering both current model capabilities and '
  'anticipated LLM-powered software.\n'
  'Our research aims to highlight the general-purpose potential of LLMs and their possible '
  'implications for US workers. Previous literature demonstrates the impressive improvements of '
  'LLMs to date (see 2.1). Our findings confirm the hypothesis that these technologies can have '
  'pervasive impacts across a wide swath of occupations in the US, and that additional '
  'advancements supported by LLMs, mainly through software and digital tools, can have significant '
  'effects on a range of economic activities. However, while the technical capacity for LLMs to '
  'make human labor more efficient appears evident, it is important to recognize that social, '
  'economic, regulatory, and other factors will influence actual labor productivity outcomes. As '
  'capabilities continue to evolve, the impact of LLMs on the economy will likely persist and '
  'increase, posing challenges for policymakers in predicting and regulating their trajectory.\n'
  'Further research is necessary to explore the broader implications of LLM advancements, '
```

Figure 4.31– Sample of result of the retrieved query

Now, our VS index is ready to be used in our chatbot. We will explore how to connect all the tools together to generate a proper human-readable answer in *Chapter 6*!

Project: Multilabel Image Classification

In the last chapter, we saved our image data into volumes. Next, we will explore our data. To follow along in your own workspace, please refer to the following notebook:

- `CH4-01-Exploring Dataset`

We create our training dataset of images. Print the labels and a sample of the training data for a look at what we want our model to classify.

```
from mlia_utils.cv_clf_funcs import *

train_df = (spark.read.format("delta").load(train_delta_path))
print(idx_class(train_df))
display(train_df)
```

▸ (5) Spark Jobs

```
{'mountain': 0, 'buildings': 1, 'glacier': 2, 'forest': 3, 'sea': 4, 'street': 5}
```

Table + New result table: OFF

	path	modificationTime	length	content
1	dbfs:/Volumes/ml_in_action/cv_clf/files/intel_image_clf/raw_images/seg_train/seg_train/buildings/0.jpg	2023-12-29T11:56:23Z	11089	
2	dbfs:/Volumes/ml_in_action/cv_clf/files/intel_image_clf/raw_images/seg_train/seg_train/buildings/10006.jpg	2023-12-29T11:56:23Z	16802	

Figure 4.32 – Load and view the training dataset